
Problem Statement: Realistic Human Face Generation using GANs with Streamlit Frontend on CelebA Dataset

Objective:

To build and train a **Generative Adversarial Network (GAN)** capable of generating **realistic human faces** using the **CelebA dataset**, and provide an **interactive Streamlit frontend** to compare real and generated images dynamically.

Description:

Generative Adversarial Networks (GANs) are a powerful class of deep learning models that can generate new data samples mimicking the distribution of real data. This project focuses on training a GAN using the **CelebA (CelebFaces Attributes) dataset** to synthesize **photorealistic human faces**.

The key outcome is not just model training but also the development of a **Streamlit-based application** where users can:

- Visualize generated faces
- Compare real vs fake samples
- Trigger model predictions dynamically through UI

This demonstrates the power of **unsupervised learning** combined with an **intuitive frontend for explainability and interaction**.

Dataset:

- **Name:** CelebFaces Attributes Dataset (CelebA)
- **Link:** [Kaggle - CelebA Dataset](#)

Download Code:

```
import kagglehub
path = kagglehub.dataset_download("jessicali9530/celeba-dataset")
print("Path to dataset files:", path)
```

- - **Contents:**
 - Over 200,000 aligned & cropped celebrity face images
 - 40 annotated binary attributes (e.g., smiling, eyeglasses, gender)
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Tools & Technologies:

- **Python, NumPy, PIL, OpenCV**
 - **TensorFlow/Keras** – for GAN architecture
 - **Streamlit** – to build interactive web UI
 - **Matplotlib** – for visualizations
 - **MLflow (optional)** – for experiment tracking and logging
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Expected Outcome:

- A trained GAN model that can generate high-quality face images.
 - A **Streamlit app** that displays:
 - Real vs generated image samples
 - On-demand generation from the GAN
 - Optionally, model training metrics or GIFs showing training progression
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